

Glasgow Footfall

1. [Navigate to Glasgow Open Data > Mobility - v1. https://developer.glasgow.gov.uk/api-details#api=mobility&operation=footfall](https://developer.glasgow.gov.uk/api-details#api=mobility&operation=footfall). This is an API for downloading hourly pedestrian count data gathered from footfall sensors in central Glasgow.
2. Click the green “Try It” button. A side panel opens. For startDate, enter a date a few months in the past in yyyy-mm-dd format, e.g., 2025-01-01. For endDate, enter today’s date.
3. At the bottom of the panel, find the “HTTP request” section. Copy the URL as shown. Note: “GET” and “HTTP/1” are not part of the URL.

The screenshot shows a web interface for testing an API. It has two main sections: 'Parameters' and 'HTTP request'.

Parameters

format	json
startDate	2025-01-01
endDate	2025-06-10

+ Add parameter

Headers

HTTP request

HTTP

Reveal secrets Copy

GET https://api.glasgow.gov.uk/mobility/v1/footfall/historical?format=json&startDate=2025-01-01&endDate=2025-06-10 HTTP/1.1

Cache-Control: no-cache

a.

4. In Power BI, navigate to the Report view > Ribbon > Home tab > Data group > Get data > Web. Paste the URL and click OK. Power Query will launch.
5. The API has generated a response in JSON format. In this case, the JSON file encodes a single table. You should find that Power Query has automatically parsed this table and produced a preview.

Table.TransformColumnTypes("#Expanded location",({"pedestrianCount", Int64.Type), ("processDate", type datetime),

	pedestrianCount	processDate	location.description	1.2 location.latitude	1.2 location.longitude
1	370	01/01/2025 00:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
2	362	01/01/2025 01:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
3	288	01/01/2025 02:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
4	303	01/01/2025 03:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
5	154	01/01/2025 04:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
6	50	01/01/2025 05:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
7	65	01/01/2025 06:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
8	64	01/01/2025 07:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
9	136	01/01/2025 08:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
10	214	01/01/2025 09:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
11	186	01/01/2025 10:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
12	405	01/01/2025 11:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
13	536	01/01/2025 12:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
14	713	01/01/2025 13:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
15	741	01/01/2025 14:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
16	913	01/01/2025 15:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
17	668	01/01/2025 16:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
18	470	01/01/2025 17:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
19	443	01/01/2025 18:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
20	387	01/01/2025 19:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
21	364	01/01/2025 20:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
22	281	01/01/2025 21:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
23	202	01/01/2025 22:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807
24	129	01/01/2025 23:00:00	Argyle Street near Queen Street Junction	55.85765	-4.252807

Query Settings

PROPERTIES

Name

historical?format=json&startDate=2025-01-01

All Properties

APPLIED STEPS

Source

Converted to Table

Expanded Column1

Expanded location

Changed Type

6. Rename the [processDate] column “processDatetime”. This will be important later if you do Optional Extension Activity 1.
7. Apply any other data cleaning or tidying that you judge appropriate. Name this Query “Glasgow Footfall”.
8. In Applied Steps, find the first Step, “Source”, and click the edit  button. Carefully modify the URL to change the start date to 2024-01-01. Leave the end date as today’s date. The screen might freeze for a minute while the API retrieves the requested data.
9. Verify that the query now returns data from the new date range. Close & Apply.
10. Create visuals to help answer the following questions.
 - a. Which location received the highest total footfall across all locations in 2024? How many pedestrians were counted? (Make sure not to include data from 2025.)
 - b. Which hour of the day received the highest total footfall across all locations in 2024?
 - c. Which location appears to be the most active at nighttime?
 - d. For location “Sauchiehall Street near Wellington Street Junction”, which month received the lowest total footfall in 2024? How many pedestrians were counted?
 - e. Which three days of the year had the lowest total footfall across all locations in 2024?
 - f. For location “Sauchiehall Street near Elmbank Street Junction”, which hour of the day received the highest total footfall in 2024? How many pedestrians were counted?
11. Save your work.

Optional Extension Activity 1

1. Clear any date filters so that your visuals now display data from 2024-01-01 to the current date, not 2024-12-31.
2. Plot the footfall day-by-day for the entire date range. Do you notice an anomaly with today's date? Why has this happened?
3. Create a Measure that returns the sum of [pedestrianCount], filtered to [processDatetime] < TODAY(). Name the Measure "pedestrianCountFilt".
4. Verify that pedestrianCountFilt hides the anomaly identified in (2).
5. Save your work.

Optional Extension Activity 2

2. Create visuals to help answer the following questions. Filter to 2024 only.
 - a. Which weekday (Mon, Tue, Wed, ...) received the most footfall overall in 2024? How many pedestrians were counted?
 - b. In 2024, which is the only location that did not receive the highest total footfall on the weekday identified in (a)? On which weekday did this location receive the highest footfall? How many pedestrians were counted?
3. Save your work.

Bristol Cycle Shops

1. Navigate to [Open Data Bristol > "Cycle shops and repairs"](https://arcg.is/0STuTL0). <https://arcg.is/0STuTL0>.
2. Download the dataset in CSV format. Import the dataset into Power BI. Name the table "Cycle Shops".
3. Inspect the [X] and [Y] columns. These are not longitude and latitude, but a different coordinate system called Ordnance Survey National Grid (OSGB), a.k.a. British National Grid. https://en.wikipedia.org/wiki/Ordnance_Survey_National_Grid. Power BI does not support OSGB coordinates in Map visuals.
4. Plot a Map visual (not Filled Map) to visualise the locations of the cycle shops. Since Power BI does not support OSGB coordinates, you will need to find a workaround using the alternative location data present in the table.
5. Are all the cycle shops present on your map and in the correct location? Find the following cycle shops on Google Maps and check whether your Power BI Map visual agrees. (Note: if you do the extension activity later, you will need to revisit these addresses, so you might want to keep those browser tabs open.)

- OBJECTID 60
Halfords - Cribbs Causeway
Centurus Retail Park, Centaurus Rd, Patchway, Bristol, BS34 5TS
- OBJECTID 2084
The Happy Bicycle Mechanic
Portishead
- OBJECTID 2890
Bristol Upcycles
Fishponds, Bristol And Bath Railway Path, Bristol, BS16 3RX

There is no need to fix any discrepancies you find. However, if this were a real-world project, we might find need to find a more robust method of associating each cycle shop with a geographic location.

6. Add a Tooltip to the Map visual and include the following fields: [NAME], [WEBSITE], [DESCRIPTION].
7. Save your work.

Optional Extension Activity 1

Let's try to plot the cycle shops more accurately on the map. We have the [X] and [Y] columns, which we know are OSGB coordinates.

1. Open Power Query and add a Custom Column to your **Cycle Shops** Query.
2. Paste the formula below. This roughly converts [X] to latitude.

$-7.896E-13 * [Y] * [Y] + 9.268E-06 * [Y] + 49.87$

3. Name this column “LatApprox”.
4. Create another Custom Column, “LonApprox”, with the following formula.

$$2.776E-11 * [X] * [X] - 5.743E-06 * [X] - 4.108$$
5. Note: the previous calculations are only approximate. It is possible to perform a more precise coordinate conversion, but the formula is complex. Let’s not do that.
6. Assign [LatApprox] and [LonApprox] the Decimal Number data type.
7. Close & Apply to get back to Power BI.
8. If your Map visual contains any fields in the “Location” well, remove them. Add [LatApprox] to the “Latitude” well and [LonApprox] to the “Longitude” well.
9. Re-check the following locations on Google Maps. Does your Power BI map now agree more closely? Are there still discrepancies?
 - OBJECTID 60
Halfords - Cribbs Causeway
Centurus Retail Park, Centaurus Rd, Patchway, Bristol, BS34 5TS
 - OBJECTID 2084
The Happy Bicycle Mechanic
Portishead
 - OBJECTID 2890
Bristol Upcycles
Fishponds, Bristol And Bath Railway Path, Bristol, BS16 3RX

Optional Extension Activity 2

1. We will now work on filtering the cycle shops by distance from Bristol City Centre. To achieve this, we will need to convert (lat, lon) differences to distances in km. We could use Power Query M language or DAX for this. For variety, let’s use DAX. On the Table view, create a new column (Ribbon > Home > Calculations group > New column). Populate it with the following formula.

```

DistFromCityCentreKm =
VAR R = 6371
VAR lat1 = RADIANS('Cycle Shops'[LatApprox])
VAR lon1 = RADIANS('Cycle Shops'[LonApprox])
VAR latCC = RADIANS(51.4538)
VAR lonCC = RADIANS(-2.60207)
VAR dLat = latCC - lat1
VAR dLon = lonCC - lon1
VAR a = 0.5 * (1 - COS(dLat) + COS(lat1) * COS(latCC) * (1 - COS(dLon)))
RETURN
    2 * R * ASIN(SQRT(a))

```

Here, (51.4538, -2.60207) are the coordinates of Bristol City Centre.

2. On the Report view, add a Slicer below the Map. Add [DistFromCityCentreKm] to the “Field” well. Verify that you can now filter by distance from the city centre.
3. Create visuals to help answer the following questions.
 - a. How many cycle shops lie within a 10 km radius of Bristol City Centre?
 - b. How many cycle shops are farther than 20 km from Bristol City Centre?
 - c. How many cycle shops are between 2 km and 5 km of Bristol City Centre?
4. Save your work.

Individual Project

1. Browse the internet and download any dataset you find that you believe might offer opportunities for creative data visualisation. A good place to start is to search for “[city name] data portal”. These data portals often provide access to many up-to-date datasets via a simple interface. Alternative sources include:
 - The Office for National Statistics
<https://www.ons.gov.uk/>
 - <https://www.data.gov.uk/>
 - Google Dataset Search
<https://datasetsearch.research.google.com/>
 - Our World in Data
<https://ourworldindata.org/>
 - NHS Digital
<https://digital.nhs.uk/>
 - Kaggle
<https://www.kaggle.com/>
Warning: many Kaggle datasets are synthetic (fake).
2. Using Power Query or other tools, clean and prepare your dataset as you judge appropriate.
3. Create a Power BI dashboard showcasing the most interesting patterns and trends in your dataset. Place an emphasis on clarity and style. Be mindful of the qualities of good data visualisation that we discussed yesterday.
4. Save your work.

Optional Extension Activity 1

1. Enhance your dashboard using a feature of Power BI you have never used before. Here are some ideas:
 - Create a custom hover-based tooltip.
<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-tooltips>
 - Create a report scheme consistent with your organisation’s branding.
<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-report-themes#customize-report-themes>
 - Experiment with the Shape Map (if not blocked by your organisation’s IT policy)
<https://learn.microsoft.com/en-us/power-bi/visuals/desktop-shape-map>
 - Set up a drillthrough page.
<https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-drillthrough>

- Generate a smart narrative for a page.
<https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-visualization-smart-narrative>

2. Save your work.

Optional Extension Activity 2

1. Repeat this project with a new dataset!

Optional Extension Activity 3

1. Repeat the project, but this time using two or more datasets that can be joined via a common column. Choose datasets that “go together” in some way, such that it would make sense to visualise them together on the same dashboard.